

## Invidivdual Power Budget

**Team Number: 306**

# Project Name: EGR 314 Subterranean Rover

**Team Member Names: Charlie, Terry, Vanessa, Keith**

**Version: 2**

| All Major Components | Component Name                                 | Part Number     | Supply Voltage Range | Qty. | Absolute Maximum Current (mA)        | Total Current (mA) | Unit |    |
|----------------------|------------------------------------------------|-----------------|----------------------|------|--------------------------------------|--------------------|------|----|
|                      | 3.3V Voltage Regulator                         | LM2575D2T-3.3R  | +4.75 - 40V          | 1    | 1000                                 | 1000               | mA   |    |
|                      | ESP32                                          | 1965-ESP32-S3-V | +3 - 3.6V            | 1    | 500                                  | 500                | mA   |    |
|                      | OLED Display                                   | ?????           | +3.3V                | 1    | 20                                   | 20                 | mA   |    |
|                      |                                                |                 |                      |      |                                      |                    |      |    |
| +3.3V Power Rail     | Component Name                                 | Part Number     | Supply               | Qty. | Absolute                             | Total              | Unit |    |
|                      | OLED Display                                   | ?????           | +3.3V                | 1    | 20                                   | 20                 | mA   |    |
|                      | ESP32                                          | 1965-ESP32-S3-V | +3 - 3.6V            | 1    | 500                                  | 500                | mA   |    |
|                      |                                                |                 |                      |      | Subtotal                             |                    | 520  | mA |
|                      |                                                |                 |                      |      | Safety Margin                        |                    | 25%  |    |
|                      |                                                |                 |                      |      | Total Current Required on +3.3V Rail |                    | 650  | mA |
|                      |                                                |                 |                      |      |                                      |                    |      |    |
| Regulator            | 3.3V Voltage Regulator                         | LM2575D2T-3.3R  | +4.75 - 40V          | 1    | 1000                                 | 1000               | mA   |    |
|                      | Total Remaining Current Available on 3.3V Rail |                 |                      |      |                                      | 350                | mA   |    |

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| <i>External Power Source 1</i>                                      | Component Name         | Part Number    | Supply    | Output | Absolute | Total | Unit |
|---------------------------------------------------------------------|------------------------|----------------|-----------|--------|----------|-------|------|
| Power Source 1 Selection                                            | USB Micro-B Connection | 670-2675-1-ND  | 5VDC      | 5VDC   | 900      | 900   | mA   |
|                                                                     |                        |                |           |        |          |       |      |
| Power Rails Connected to External Power Source 1                    |                        |                |           |        |          |       |      |
|                                                                     | +3.3V Regulator        | LM2575D2T-3.3R | +5V - 20V | 1      | 500      | 500   | mA   |
| <b>Total Remaining Current Available on External Power Source 1</b> |                        |                |           |        |          | 400   | mA   |

**Notes:**

External Supply Voltage should be determined by the dropout voltage for highest-voltage regulator (e.g., +14V for a +12V regulator).  
 If you have multiple units in your design (e.g., a base unit and remote unit) then you need a separate power budget for each unit